

▲ Figure 9.6 Metals are malleable because the layers of positive ions can slide over each other without changing the bonding.

Metals can also be described as **ductile**. This means that they can be drawn out into wires. The explanation is the same as why they are malleable.

CHAPTER QUESTIONS

SKILLS INTERPRETATION 7

1 Draw simple 2D diagrams to show the structure of:
- a potassium metal
- b calcium metal

SKILLS CRITICAL THINKING 8

SKILLS REASONING 10

2 a A solid metal is often described as having 'a lattice of positive ions in a sea of delocalised electrons'. State the electronic configuration of a magnesium atom and use it to explain what this phrase means.

b Metallic bonds are not fully broken until the metal has first melted and then boiled. Sodium, magnesium and aluminium are three consecutive elements in the Periodic Table. The boiling points of sodium, magnesium and aluminium are 890 °C, 1110 °C and 2470 °C, respectively. Explain these values in terms of the electronic configurations of the elements and metallic bonding.

7 c Explain why all these elements are good conductors of electricity.

d Explain why all these metals are malleable.

9 3 This question uses ideas from Chapters 7, 8 and 9. In these chapters you have met the following types of structure and bonding:

giant metallic structure
giant covalent structure
giant ionic structure
covalent molecular structure

Some information about some substances is given below. In each case state what type of structure and bonding it has.

a Substance A melts at 2300 °C. It doesn't conduct electricity even when it is molten. It is insoluble in water.

8 b Substance B is a colourless gas.

9 c Substance C is a yellow solid with a low melting point of 113 °C. It doesn't conduct electricity and it is insoluble in water.

d Substance D forms brittle orange crystals which melt at 398 °C. It dissolves freely in water to give an orange solution.

e Substance E is a pink-brown flexible solid. It conducts electricity.

f Substance F is a liquid with a boiling point of 80 °C. It is insoluble in water.

g Substance G is a silvery solid which melts at 660 °C. It is used in overhead power cables.

END OF CHEMISTRY ONLY